

Q7a) Explain the algorithm for Classical Planning with an example.

Classical Planning is an AI technique used to find a sequence of actions that transforms an initial state into a goal state.

It assumes: Environment is fully observable, Deterministic actions, Static environment, Finite actions

Components of Classical Planning - Initial State, Goal State, Actions/Operators, Preconditions, Effects

Planning Algorithm - The planning algorithm searches for sequence of actions that satisfy the goal.

Steps of Classical Planning Algorithm -

Step 1: Define Initial State - Current state of environment.

Step 2: Define Goal State - Desired final state.

Step 3: Define Actions - Each action contains: Preconditions, Effects

Step 4: Select Applicable Action - Choose action whose preconditions are satisfied.

Step 5: Apply Action - Update state according to action effects.

Step 6: Repeat - Continue until goal state is achieved.

General Planning Algorithm -

1. Start with initial state
2. Compare with goal state
3. Select applicable action
4. Apply action
5. Generate new state
6. Repeat until goal reached

Example: Robot Planning Problem

Initial State - Robot at Room A, Box at Room B

Goal State - Box at Room A

Actions -

Move(x,y) -

Preconditions - Robot at x

Effect - Robot moves to y

Pick(Box) -

Preconditions - Robot and box at same room

Effect - Robot holds box

Drop(Box) -

Preconditions - Robot holding box

Effect - Box placed in room

Solution Plan -

1. Move(RoomA,RoomB)
2. Pick(Box)
3. Move(RoomB,RoomA)
4. Drop(Box)

Goal achieved: Box reaches Room A.

Advantages - Systematic problem solving, Goal-oriented, Useful in robotics

Applications - Robotics, Automated scheduling, Game AI, Navigation systems.

Q7b) Analyze various planning approaches in AI.

Planning is the process of generating sequence of actions to achieve goals. Different planning approaches are used depending on problem complexity.

Various Planning Approaches in AI

1. State Space Planning
2. Goal Stack Planning
3. Hierarchical Planning
4. Nonlinear Planning
5. Conditional Planning
6. Reactive Planning

1. State Space Planning - Searches through states to reach goal. Two types: Forward planning & Backward planning.

Forward Planning - Starts from initial state and moves toward goal. Advantages - Simple, Easy implementation. Disadvantages - Explores many states

Backward Planning - Starts from goal and works backward. Advantages - Efficient for specific goals. Disadvantages - Complex reasoning

2. Goal Stack Planning - Uses stack of goals and subgoals. AI solves one subgoal at a time. Advantages - Structured planning, Easy decomposition

3. Hierarchical Planning - Breaks large problem into smaller tasks. Uses hierarchy of goals. Advantages - Reduces complexity, Efficient planning

4. Nonlinear Planning - Actions are partially ordered instead of fixed sequence.

Allows parallel execution. Advantages - Flexible planning, Parallel actions possible

5. Conditional Planning - Used when environment is uncertain. Plan depends on conditions. Example - "If raining then carry umbrella."

6. Reactive Planning - Agent reacts immediately to environment changes. No long-term planning required. Applications - Robotics, Autonomous vehicles

Q7c) Explain Hierarchical Planning with relevant example.

Hierarchical Planning is a planning approach in which a large problem is divided into smaller subproblems arranged in hierarchy. Also called: Hierarchical Task Network (HTN) Planning.

Main Idea - Complex tasks are decomposed into: High-level tasks, Subtasks, Primitive actions

Steps in Hierarchical Planning

1. Define main goal
2. Divide into subtasks
3. Further decompose subtasks
4. Execute primitive actions

Structure -

Example: Travel Planning

Goal - Go to College

Decomposition -

Go to College
↓
Get Ready
↓
Wear Uniform
Pack Bag
Eat Breakfast
↓
Travel
↓
Take Bus
Reach College

Main Goal
↓
Subtasks
↓
Primitive Actions

Another Example: Robot Planning

Goal: Clean Room

Subtasks: Pick objects, Arrange furniture, Sweep floor

Primitive actions: Move, Pick, Drop

Advantages of Hierarchical Planning - Reduces complexity, Easy problem decomposition, Faster planning, Reusable subtasks

Disadvantages - Requires domain knowledge, Difficult hierarchy design

Applications - Robotics, Game AI, Automated assistants, Industrial automation

Hierarchical Planning simplifies complex problems by dividing them into smaller manageable tasks, making AI planning more efficient and organized.

Q8a) Explain with example how Planning is different than Problem Solving.

Problem solving is the process of finding a sequence of actions to move from: Initial state to Goal state using search algorithms.

Examples: Puzzle solving, Path finding

Planning is a goal-directed reasoning process where AI determines: What actions to perform, In what order, To achieve desired goal

Planning considers: Preconditions, Effects of actions

Difference between Planning and Problem Solving

Problem Solving	Planning
Uses search techniques	Uses action reasoning
Focuses on finding path	Focuses on action sequence
Less knowledge about actions	Uses detailed action knowledge
State-based search	Goal-based reasoning
Suitable for simple problems	Suitable for complex real-world tasks
Example: Maze solving	Example: Robot task planning

Q8c) Explain AI Components and AI Architecture.

Artificial Intelligence (AI) is the branch of computer science that enables machines to perform intelligent tasks such as: Learning, Reasoning, Decision making

Components of AI -

1. **Knowledge Base** - Stores: Facts, Rules, Information

Example: Human(Ram)

Human(x) → Mortal(x)

2. **Inference Engine** - Applies logical reasoning on knowledge base to derive conclusions. Uses: Forward chaining, Backward chaining

3. **Learning Component** - Allows system to learn from: Data, Experience, Environment.
Example: Machine learning systems

4. **Perception Component** - Collects information using sensors. Examples: Camera, Microphone

5. **Planning Component** - Determines sequence of actions to achieve goals.

6. **Natural Language Processing (NLP)** - Enables communication between humans and machines. Example: Chatbots, Voice assistants

7. **Actuators** - Perform physical actions. Examples: Robot arms, Motors

AI Architecture defines structure and interaction of AI components.

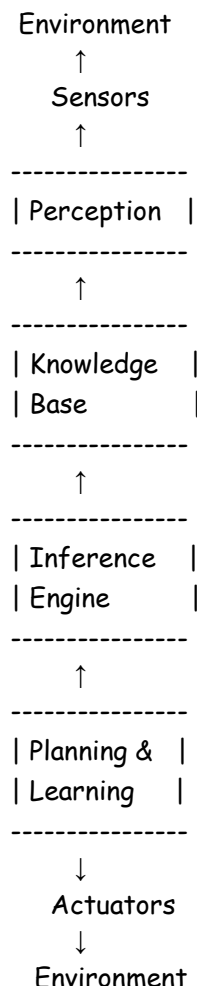
Basic AI Architecture -

Working of AI Architecture -

1. Sensors collect data.
2. Perception interprets data.
3. Knowledge base stores information.
4. Inference engine performs reasoning.
5. Planning module selects actions.
6. Actuators execute actions.

Applications of AI Architecture - Robotics,
Self-driving cars, Expert systems, Virtual assistants,
Healthcare AI.

Advantages - Intelligent decision making, Automation,
Efficient reasoning, Learning capability.



Q7a) Describe the differences and similarities between Problem Solving and Planning.

Similarities -

- Both aim to achieve a goal
- Both use search techniques
- Both involve states and actions
- Both require reasoning
- Both are used in intelligent agents

Problem Solving Example - Finding shortest path between two cities using BFS.

Planning Example - Robot making tea:

1. Boil water
2. Add tea powder
3. Add sugar
4. Serve tea

Each action has: Preconditions, Effects

Problem solving mainly searches for paths, whereas planning reasons about actions and their consequences to achieve goals efficiently.

Q7c) What are the different types of planning? How planning algorithm can be represented as State Space Search?

Types of Planning in AI ...

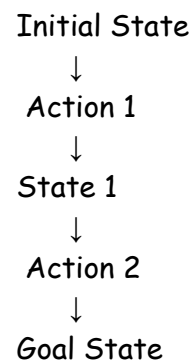
Planning methods differ according to: Complexity, Environment, Goal requirements

Planning as State Space Search - Planning problems can be represented as state space search where:

- States → Configurations of environment
- Actions → Operators changing states
- Goal test → Checks whether goal achieved

Components - Initial State, Goal State, Actions, Transition Model

Representation -



Example - Robot planning problem:

Initial State - Robot at Room A & **Goal State** - Box at Room A

Actions - Move, Pick, Drop

Solution Path -

Move(A,B)

Pick(Box)

Move(B,A)

Drop(Box)

Goal achieved through state transitions.

Advantages of State Space Planning - Systematic search, Goal-oriented reasoning, Suitable for robotics

Q8b) Explain The Blocks World in detail.

Blocks World is a classical AI planning problem used to study: Planning, Reasoning, Problem solving.

It consists of: A table, Several blocks.

The goal is to arrange blocks into desired configuration.

Components of Blocks World - Blocks, Table, Robot Arm

Operations in Blocks World -

Operation	Meaning
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PICKUP(x)	Pick block x
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PUTDOWN(x)	Put block on table
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STACK(x,y)	Put x on y
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UNSTACK(x,y)	Remove x from y
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Conditions Used -	Predicate	Meaning
	ON(x,y)	x is on y
	ONTABLE(x)	x is on table
	CLEAR(x)	No block on x
	HOLDING(x)	Robot holding x
	ARMEMPTY	Robot arm empty

Example -

Initial State

A on table

B on table

C on A

Goal State

B on A

C on B

Planning Steps -

1. UNSTACK(C,A)

2. PUTDOWN(C)

3. PICKUP(B)

4. STACK(B,A)

5. PICKUP(C)

6. STACK(C,B)

Goal achieved.

Applications - Robotics, Automated planning, AI learning systems

Advantages - Simple planning model, Useful for AI research, Demonstrates goal-based planning.

Q8c) What are the limitations of AI? Explain future scopes with AI.

Limitations of AI -

1. **Lack of Human Intelligence** - AI cannot fully replicate: Creativity, Emotions, Common sense
2. **High Development Cost** - AI systems require: Expensive hardware, Large datasets, Skilled experts
3. **Dependency on Data** - Poor or biased data leads to incorrect decisions.
4. **No Emotional Understanding** - AI cannot understand: Feelings, Human emotions
5. **Limited Reasoning Ability** - AI works only within trained domains.
6. **Job Displacement** - Automation may reduce human employment opportunities.
7. **Security and Privacy Issues** - AI systems may misuse personal data.

Future Scope of AI -

1. **Healthcare** - Disease prediction, Medical diagnosis, Robotic surgery
2. **Education** - Smart tutoring systems, Personalized learning
3. **Robotics** - Industrial robots, Service robots
4. **Transportation** - Self-driving cars, Smart traffic systems
5. **Agriculture** - Crop monitoring, Smart irrigation
6. **Cybersecurity** - Threat detection, Fraud prevention
7. **Business and Finance** - Chatbots, Market prediction, Automation
8. **Space Research** - Autonomous exploration, Intelligent spacecraft

Advantages of Future AI - Increased automation, Faster decision making, Improved productivity, Better healthcare systems

Although AI has certain limitations, its future scope is enormous and it will continue transforming industries such as healthcare, education, transportation, robotics, and business.

Q7a) Explain with an example Goal Stack Planning (STRIPS Algorithm).

Goal Stack Planning is a planning technique in AI that solves problems by dividing goals into smaller subgoals using a stack data structure.

It is mainly based on: STRIPS (Stanford Research Institute Problem Solver)

STRIPS is a classical planning language used to represent: Initial state, Goal state, Actions with: Preconditions & Effects

Components of STRIPS -

1. **Initial State** - Current state of environment.
2. **Goal State** - Desired final state.
3. **Operators / Actions** - Each action contains:

Preconditions - Conditions that must be true before action.

Add List - Facts added after action.

Delete List - Facts removed after action.

Goal Stack Planning Algorithm - The planner uses stack to store: Goals, Subgoals, Actions

Steps of Goal Stack Planning -

1. Push goal onto stack
2. If goal satisfied → remove it
3. Otherwise select action achieving goal
4. Push action and its preconditions
5. Solve subgoals recursively
6. Execute action
7. Repeat until stack empty

Example: Blocks World

Initial State - ON(A,Table)
ON(B,Table)

Goal State - ON(A,B) CLEAR(A)
CLEAR(B)
Need to place A on B. ARMEMPTY

STRIPS Operator - STACK(A,B)

Preconditions - HOLDING(A)
CLEAR(B)

Add List - ON(A,B)
ARMEMPTY

Delete List - HOLDING(A)
CLEAR(B)

Planning Process -

Step 1 - Push Goal: ON(A,B)

Step 2 - Select action: STACK(A,B)

Push Preconditions: HOLDING(A)
CLEAR(B)

Step 3 - CLEAR(B) already true.

Need: HOLDING(A)

Select action: PICKUP(A)

Preconditions: ON(A,Table)
CLEAR(A)
ARMEMPTY

All are true.

Final Plan -

1. PICKUP(A)
2. STACK(A,B)

Goal achieved: $ON(A,B)$

Advantages of Goal Stack Planning - Goal-oriented planning, Efficient decomposition of problems, Uses subgoals systematically

Disadvantages - Difficult for large complex domains, May generate redundant subgoals

Applications - Robotics, Automated planning, Game AI

Q8a) Explain Planning in Non-Deterministic Domain.

Non-Deterministic Planning - A non-deterministic domain is an environment where actions may produce multiple possible outcomes.

The exact result of an action is uncertain.

Example - Action: Throwing a ball

Possible outcomes: Hits target, Misses target

Need for Non-Deterministic Planning -

Classical planning assumes: One action $\rightarrow$ One predictable result

But real-world environments contain: Uncertainty, Incomplete information, Random events, Hence non-deterministic planning is needed.

Characteristics - Multiple outcomes possible, Uncertain environment, Conditional actions required, Flexible planning needed

Types of Planning in Non-Deterministic Domain -

1. **Conditional Planning** - Plan depends on conditions.

Example - If raining $\rightarrow$ Carry umbrella
Else $\rightarrow$ Do not carry umbrella

2. **Contingency Planning** - Creates backup plans for different situations.

Example - If road blocked: take alternate route

3. **Probabilistic Planning** - Uses probabilities to handle uncertain outcomes.

Planning Representation -

State + Action

$\downarrow$

Multiple Possible States

Example: Robot Navigation

Robot tries to move forward.

Possible outcomes: Moves correctly, Slips, Obstacle encountered

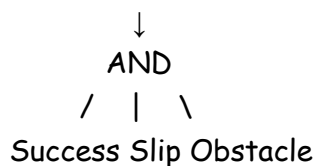
Planner must prepare actions for all outcomes.

AND-OR Search Tree - Non-deterministic planning often uses: AND nodes, OR nodes

OR Node - Represents choice of action.

AND Node - Represents all possible outcomes of action.

Example - Move Forward



Planner must handle all branches.

Advantages - Handles uncertainty, More realistic planning, Useful for dynamic environments

Disadvantages - Complex planning, High computational cost, Difficult reasoning

Applications - Robotics, Self-driving cars, Autonomous agents, Medical diagnosis systems

Planning in non-deterministic domains helps AI systems operate effectively in uncertain and unpredictable environments by preparing for multiple possible outcomes.

Q7a) Write a short note on Planning Agent, State Goal and Action Representation.

Planning Agent - A Planning Agent is an intelligent agent that determines a sequence of actions to achieve a specific goal.

It: Perceives environment, Generates plan, Executes actions, Achieves goal efficiently

Working of Planning Agent

1. Observe current state
2. Compare with goal state
3. Generate action sequence
4. Execute plan
5. Reach desired goal

Environment



Sensors



Planning Agent



Planner



Action Sequence



Actuators



Environment

Architecture of Planning Agent -

Components of Planning Agent - Knowledge Base, Planner, Goal Representation, Action Representation, Execution Module

State Representation - A state describes the condition of environment at a particular time.

Example - In Blocks World:

ON(A,B)
CLEAR(A)
ONTABLE(B)

This represents current arrangement of blocks.

Goal Representation - Goal representation specifies desired final state to be achieved.

Example - ON(A,B)

Meaning: Block A should be on block B.

Action Representation - Actions describe operations that change states.

Each action contains: Preconditions, Effects

Example - STACK(A,B)

Preconditions - HOLDING(A)
CLEAR(B)

Effects - ON(A,B)
ARMEMPTY

Advantages of Planning Agent - Goal-oriented behavior, Efficient decision making, Useful in robotics and automation

Applications - Robotics, Navigation systems, Game AI, Automated scheduling

Q7b) Explain different components of Planning System.

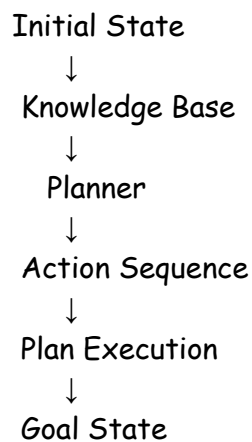
A planning system generates sequence of actions to achieve a goal from initial state.

Components of Planning System -

1. **Initial State** - Represents current condition of environment. Example - Robot at Room A
2. **Goal State** - Represents desired condition to achieve. Example - Robot at Room B
3. **Actions / Operators** - Actions used to move from one state to another. Each action contains: Preconditions, Effects.

- 4. **State Space** - Collection of all possible states.
- 5. **Planner** - Core component that: Selects actions, Generates plans, Searches solution path
- 6. **Knowledge Base** - Stores: Facts, Rules, Constraints
- 7. **Plan Executor** - Executes generated actions in environment.
- 8. **Environment** - Real world where agent performs actions.

Diagram of Planning System



Applications - Robotics, Industrial automation, AI game systems, Autonomous vehicles

Q8a) Explain the terms for Time and Schedule from perspective of Temporal Planning.

Temporal Planning is a planning method where actions have: Time duration, Scheduling constraints

It considers:

1. Start time
2. End time
3. Action duration
4. Resource allocation

Time in Temporal Planning - Time represents duration and ordering of actions.

Actions may: Start at different times, Execute simultaneously, Depend on time constraints

Example - Action: Travel
Duration: 2 hours

Features of Time Representation - Action duration, Time constraints, Concurrent actions, Deadlines

Schedule in Temporal Planning - Schedule specifies: When actions should occur, Order of actions, Resource usage

Example Schedule -

Action	Start Time	End Time
Prepare Tea	8:00 AM	8:10 AM
Serve Tea	8:10 AM	8:15 AM

Types of Scheduling Constraints -

1. Precedence Constraints - One action before another
2. Resource Constraints - Limited resources
3. Time Constraints - Deadline or fixed duration

Temporal Planning Representation -

Action + Duration + Constraints



Schedule

Example - Robot Task:

1. Pick object (2 min)
2. Move object (5 min)
3. Place object (1 min)

Total schedule created considering time durations.

Advantages of Temporal Planning - Handles real-world timing constraints, Supports concurrent actions, Efficient resource management

Applications - Robotics, Space missions, Manufacturing systems, Project scheduling, Autonomous vehicles

Temporal planning combines planning with time and scheduling constraints, enabling AI systems to perform complex real-world tasks efficiently.